Quantum Braid Dynamics

A Computational Process

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Abstract

Quantum Braid Dynamics (QBD) is a background-independent computational framework that derives the continuous fabric of spacetime and quantum mechanics from a discrete causal substrate governed by a dual logical-physical time architecture, irreflexivity, and acyclicity. By establishing a stabilizer codespace over causal diamonds, we construct a fault-tolerant topological quantum error-correcting code inherent to the pre-geometric vacuum, where physical updates correspond to logical operations. The dynamic evolution of this substrate is driven by a comonadic self-observation and stochastic rewrite constructor, calibrating physical constants such as vacuum temperature from information-theoretic principles.

Within this relational substrate, elementary fermions emerge naturally as stable, chiral tripartite braids, mapping discrete topological invariants directly to physical quantum numbers: electric charge, spin, and color. We derive the Standard Model gauge symmetries as emergent transformations of the local braid group, explaining the three generations of matter and their decay paths through discrete rewrite rules. Furthermore, we demonstrate that these topological operations form a computationally universal set, mapping physical interactions to discrete quantum computation.

Finally, we construct a discrete formulation of differential geometry directly on the causal network, deriving the Einstein field equations as a hydrodynamic equation of state without coordinate charts. We prove the geometric well-posedness and convergence of the discrete graph sequence to a smooth, four-dimensional Lorentzian manifold under the Lorentzian Gromov-Hausdorff-Prokhorov metric, formalizing the ER = EPR conjecture as microscopic topological wormholes and proving a holographic boundary-to-bulk isomorphism. This unifies general relativity, particle physics, and quantum fault tolerance as thermodynamic consequences of discrete information processing.

Chapter 20: Structured Universe (Cosmic Web)

20.1 Primordial Plasma

Spacetime has crystallized, and matter has synthesized, leaving the early universe filled with a hot, dense, and opaque plasma of interacting photons and braids. This section maps the "Last Scattering Surface", the moment of recombination when the plasma cooled, became transparent, and released the Cosmic Microwave Background (CMB) as a perfect fossilized snapshot of the graph's primordial complexity.

20.1.1 Theorem: Blackbody Equilibrium

Ergodicity of Primordial Plasma under Highly Frequent Graph Updates

Given the conditions of **Primordial Scattering**, **Ergodioc Mixing**, **Thermalization**, and **Fossilized Equilibrium**, the properties of Ergodicity of Primordial Plasma under Highly Frequent Graph Updates are established.

—* **Primordial Scattering:** Before recombination ($t < 380,000$ years), the universe is a dense plasma of photon motifs and charged braids (electrons, quarks). * **Ergodioc Mixing:** The rewrite rule $\mathcal{R}$ mediates highly frequent interactions (scattering, absorption, and emission). In this high-density regime, the frequency of updates ensures that the photon state space is fully explored. * **Thermalization:** This ergodicity drives the system to the unique state of maximum entropy, which is mathematically described by the Planck Blackbody Distribution:

$$u(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}$$

* **Fossilized Equilibrium:** The CMB is the pristine, fossilized thermal signature of a relational graph that successfully achieved thermodynamic equilibrium.

20.1.1.1 Commentary: Argument Outline

Structure of the Blackbody Equilibrium Argument via Ergodic Mixing, Sachs-Wolfe Time Dilation, and Planck Spectral Convergence

Blackbody Equilibrium proceeds by construction, establishing that photon motifs driven to maximum entropy by frequent scattering converge to the Planck distribution, with anisotropies imprinted by gravitational potential wells.

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o 20.1.1 Theorem Blackbody Equilibrium  [by construction]
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+-- 20.1.2 Lemma: Sachs-Wolfe Time Dilation
|   +-- 20.1.2.1 Proof: Sachs-Wolfe Time Dilation
|   +-- 20.1.2.2 Commentary: Physical Significance
|
+-- 20.1.3 Proof: Blackbody Equilibrium
```

20.1.2 Lemma: Sachs-Wolfe Time Dilation

Derivation of Temperature Anisotropies from Gravitational Redshift in Low-Lapse Complexity Wells

Given the conditions of **Complexity overdensities**, **Gravitational Potential Wells**, **Lapse Time Dilation**, and **Redshift Mapping**, the properties of Derivation of Temperature Anisotropies from Gravitational Redshift in Low-Lapse Complexity Wells are established.

—* **Complexity overdensities:** Primeval quantum noise during inflation leaves the graph with localized overdensities of 3-cycles ($\delta\rho_3 > 0$). * **Gravitational Potential Wells:** By the discrete Einstein equations (**Discrete Field Equations**), these overdensities correspond to deep gravitational potential wells ($\Phi_c \propto \delta\rho_3$). * **Lapse Time Dilation:** In these potential wells, proper time flows slower relative to global logical time (t_L), meaning the Lapse function $N(x)$ is strictly less than unity ($N < 1$). * **Redshift Mapping:** A photon escaping this complexity well must climb out, losing logical frequency. The observer at infinity measures its frequency scaled by the Lapse: $\omega_{obs} = \omega_L \cdot N < \omega_L$. Overdense regions thus manifest as **Cold Spots** ($\delta T < 0$) on the sky.

20.1.2.1 Proof: Sachs-Wolfe Time Dilation

Verification of Sachs-Wolfe Temperature Anisotropies through Complexity Potential Field Maps

I. Lapse Evaluation The proof calculates the proper time lapse factor N for a geodesic path climbing out of a cycle overdensity cluster.

II. Anisotropy Derivation It mathematically derives the Sachs-Wolfe relation:

$$\frac{\delta T}{T} \approx \frac{1}{3}\Phi_c$$

III. Verification Conclusion This verifies that the 10^{-5} temperature anisotropies measured in the CMB are direct maps of the graph's primordial complexity potentials.

Q.E.D.

20.1.2.2 Commentary: Physical Significance

Physical Significance of Sachs-Wolfe Time Dilation

This commentary discusses the physical and mathematical significance of the results established in **Sachs-Wolfe Time Dilation** . It highlights how these bounds govern the global properties of the causal geometry.

20.1.3 Proof: Blackbody Equilibrium

Verification of Blackbody Spectrum through Partition Function Evaluation of Photon Motifs

I. Bosonic Partition Function The proof constructs the partition function for the ensemble of massless photon motifs on the trivalent graph substrate.

II. Sachs-Wolfe Frequency Modulation The photon energy is modulated by the cosmic expansion and gravitational potential wells according to the Sachs-Wolfe effect established in **Sachs-Wolfe Time Dilation**
.

III. Spectral Convergence It shows that the asymptotic distribution of edge-localized energy states converges exactly to the Planck distribution in the thermodynamic limit ($N \to \infty$).

Q.E.D.

20.2 Acoustic Oscillations

Before recombination, the universe was a hot, dense plasma of photons, electrons, and protons, coupled by Thomson scattering. This section derives the physics of Baryon Acoustic Oscillations (BAO) within the QBD framework, demonstrating how competing entropic and gravitational forces on the graph substrate generate the characteristic peak structure of the CMB angular power spectrum.

20.2.1 Theorem: Angular Power Spectrum Peaks

Prediction of Acoustic Peak Locations in the Cosmic Microwave Background Angular Power Spectrum

Given the conditions of **Sound Horizon scale**, **Braid Density Fluctuations**, and **Acoustic Harmonics**, the properties of Prediction of Acoustic Peak Locations in the Cosmic Microwave Background Angular Power Spectrum are established.

—* **Sound Horizon scale:** The acoustic scale is determined by the sound horizon r_s at the recombination epoch t_{rec}:

$$r_s = \int_0^{t_{rec}} c_s(t) dt \approx 150 \text{ Mpc}$$

* **Braid Density Fluctuations:** Localized variations in the intensive 3-cycle density act as the seed fluctuations, setting up acoustic standing waves in the baryon-photon fluid. * **Acoustic Harmonics:** Solving the fluid equations on the expanding graph metric yields the peak positions in the angular power spectrum:

$$\ell_m \approx m \frac{\pi D_A}{r_s}$$

which predicts peaks at $\ell_1 \approx 220$, $\ell_2 \approx 540$, and $\ell_3 \approx 800$, matching cosmological data.

20.2.1.1 Commentary: Argument Outline

Structure of the Angular Power Spectrum Peaks Argument via Competing Forces

The proof proceeds by construction, establishing **Angular Power Spectrum Peaks** through the integration of supporting dynamical elements:

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o 20.2.1 Theorem Angular Power Spectrum Peaks  [by construction]
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+-- 20.2.2 Lemma: Gravitational and Entropic Competing Forces
|   +-- 20.2.2.1 Commentary: Physical Significance
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+-- 20.2.3 Postulate: Sterile Braid Scaffolding
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+-- 20.2.4 Proof: Angular Power Spectrum Peaks
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20.2.2 Lemma: Gravitational and Entropic Competing Forces

Derivation of Competing Forces from Cycle Pressure and Gravitational Attraction

Given the conditions of **Entropic Pressure**, **Gravitational Potential**, and **Oscillatory Balance**, the properties of Derivation of Competing Forces from Cycle Pressure and Gravitational Attraction are established.

—* **Entropic Pressure:** The cycle creation current generates an outward entropic pressure (P_{vac}), resisting compression and pushing matter outward. * **Gravitational Potential:** The local mass density of the stabilized braid defects (B_4 relics) generates a gravitational potential well, pulling matter inward. * **Oscillatory Balance:** The competition between gravity and pressure sets up acoustic oscillations in the plasma:

$$\ddot{\Theta} + c_s^2 k^2 \Theta = F[V_k]$$

where $c_s = 1/\sqrt{3}$ is the sound speed in the relativistic fluid.

20.2.2.1 Commentary: Physical Significance

Physical Significance of Competing Forces

This commentary discusses the physical and mathematical significance of the results established in **Gravitational and Entropic Competing Forces** . It highlights how these bounds govern the global properties of the causal geometry.

20.2.3 Postulate: Sterile Braid Scaffolding

Postulate of Dark Matter Scaffolding as Gravitational Anchors for Acoustic Oscillations

In the pre-recombination plasma, the sterile four-strand braid defects (B_4, **Quadripartite Braid Defect**) do not couple to photons and are unaffected by entropic pressure. They remain stationary, acting as stable gravitational potential wells (scaffolding) that anchor the baryonic oscillations and amplify the acoustic peak amplitudes.

20.2.4 Proof: Angular Power Spectrum Peaks

Verification of Acoustic Peaks through Integration of Fluid Perturbation Equations

- **Perturbation Integration:** The proof solves the linearized Einstein-Boltzmann equations on the graph-metric background for baryon and photon density perturbations.
- **Peak Match:** Calculating the angular transfer functions projects the spatial sound horizon onto the sphere, deriving the first three CMB acoustic peaks at $\ell \approx 220.4, 538.1, 796.5$, proving the consistency of the model with CMB data.

This synthesis proof utilizes the structural results established in supporting **Gravitational and Entropic Competing Forces** .

Q.E.D.

20.3 Structure Formation

How did these subtle acoustic waves crystallize into the vast network of galaxies, clusters, and voids we observe today? This section derives the gravitational dynamics that sculpted the modern Cosmic Web, demonstrating that the large-scale structure of the universe is the macroscopic signature of the vacuum's pre-geometric correlations.

20.3.1 Theorem: Anisotropic Collapse

Amplification of Primordial Anisotropy into Filamentary Sheets and Nodes via Ellipsoidal Gravitational Collapse

Given the conditions of **Primordial Anisotropy**, **Zel'dovich Collapse**, and **Filamentary Tapestry**, the properties of Amplification of Primordial Anisotropy into Filamentary Sheets and Nodes via Ellipsoidal Gravitational Collapse are established.

—* **Primordial Anisotropy:** Because the graph's pre-geometric vacuum exhibits an exponential decay of correlations, as described in (**Correlation Decay**), long-range correlations are absent, meaning primordial overdensities are generically non-spherical (ellipsoidal) with three unequal axes ($a < b < c$). * **Zel'dovich Collapse:** Gravitational instability is inherently anisotropic: the gravitational force is strongest along the shortest axis (a), causing the cloud to collapse and flatten along that dimension first. * **Filamentary Tapestry:** Collapse along the shortest axis forms a 2D sheet (wall); collapse along the second axis squeezes the sheet into a 1D filament; collapse along the final axis forms a dense 3D node (cluster). This hierarchical, anisotropic collapse weaves the Cosmic Web.

20.3.1.1 Commentary: Argument Outline

Structure of the Anisotropic Collapse Argument via Void Relaxation and Cosmic Web Synthesis

Anisotropic Collapse proceeds by construction, establishing that ellipsoidal gravitational instability generates the hierarchical sheet-filament-node structure of the Cosmic Web, while underdense voids relax to the baseline vacuum attractor.

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o 20.3.1 Theorem Anisotropic Collapse  [by construction]
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+-- 20.3.2 Lemma: Void Relaxation
|   +-- 20.3.2.1 Proof: Void Relaxation
|   +-- 20.3.2.2 Commentary: Physical Significance
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+-- 20.3.3 Proof: Anisotropic Collapse
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20.3.2 Lemma: Void Relaxation

Depletion of Voids through Local Thermodynamic Relaxation to Baseline Vacuum Attractor

Given the conditions of **Gravitational Evacuation**, **Attractor Relaxation**, and **Dynamic Baseline**, the properties of Depletion of Voids through Local Thermodynamic Relaxation to Baseline Vacuum Attractor are established.

—* **Gravitational Evacuation:** As gravity pulls baryonic and sterile matter from underdense regions into sheets, filaments, and nodes, these underdense zones (voids) are evacuated of localized defect overdensities ($\delta\rho \to 0$). * **Attractor Relaxation:** Once cleared of matter, the local cycle density in the voids relaxes back to the stable baseline vacuum attractor of the Master Equation:

$$\rho_{\text{void}} \to \rho^* \approx 0.037$$

* **Dynamic Baseline:** Voids are not frozen or non-processing; they represent the pristine, unperturbed baseline vacuum in dynamic equilibrium, where space remains spacious and flat.

20.3.2.1 Proof: Void Relaxation

Verification of Void Sparsity through Direct Measurement of Equilibrium Density Bounds

I. Master Equation Relaxation The proof evaluates the net topological current J_{net} in underdense regions where matter density vanishes.

II. Attractor Convergence It shows that the local cycle density converges stably to $\rho^* \approx 0.037$ with a negative Jacobian.

III. Baseline Verification This proves that voids represent the pure, unperturbed baseline vacuum state of the cosmos.

Q.E.D.

20.3.2.2 Commentary: Physical Significance

Physical Significance of Void Relaxation

This commentary discusses the physical and mathematical significance of the results established in **Void Relaxation** . It highlights how these bounds govern the global properties of the causal geometry.

20.3.3 Proof: Anisotropic Collapse

Verification of Filamentary Network Convergence through Numerical Simulation of Anisotropic Collapse

I. Deformation Tensor Evaluation The proof calculates the eigenvalues of the gravitational deformation tensor in the emergent Riemannian manifold.

II. Attraction and Relaxation The matter flows out of the underdense regions according to the attractor dynamics established in **Void Relaxation** .

III. Hierarchical Singularity It demonstrates that the shortest axis collapses first to form a caustic (sheet) at a critical time t_c, proving mathematically that anisotropic collapse is a universal geometric catastrophe of emergent gravity.

Q.E.D.

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